

Intelligent Decisions and the Brain

Alan Turing's central insight in suggesting his famous test of intelligence is that it is the nature and quality of the output that matters. Unfortunately, modern versions of this test, e.g., the Loebner prize, nicely prove the maxim that any measurement becomes hopelessly corrupted when used as a metric. To read the transcripts of the conversations between judges, humans and computer programmes is to experience a ratcheting ridiculousness. Programmes produce fatuous utterances in order to be taken for humans, who in turn produce even more preposterous prattle to try to escape. It is a curious irony indeed that intelligence should come to be evaluated according to nonsense.

Much of the problem with Turing's test is that it results in a language game, and it has been known from the days of ELIZA that we have an anthropomorphizing soft spot for semantically vacuous verbals. It would surely be better instead to assess the intelligence of decisions or choices – and indeed this is our starting point. We wish to understand and replicate the capacity to make robust decisions in circumstances of little information and rampant dangers, and to learn and generalize appropriately from success and failure. Fortunately, this capacity plays a key part in a swathe of technical fields, including control theory, economics, machine learning, operations research and particularly reinforcement learning, so we have giants on whose shoulders we can stand.

Another aspect of Turing's agenda is that mechanism does not matter. That is, we can happily ignore the myriad details of how humans actually produce their intelligent decisions. Although the conclusion is formally true, many scholars have pointed out the problems of straying too far from our single best test cases for intelligence. It turns out that we are in a golden age for decision making, with insights and ideas from the technical fields cross-fertilizing seamlessly with psychological data on human and animal behaviour, and neural data on systems and mechanisms.

There are of course many too many topics to cover. However, one primary insight from this convergence is the existence of multiple cooperating and competing decision-making systems. Decision-making is hard since it is typically necessary to worry about whole sequences of choices rather than single ones. An early insight in artificial intelligence from Arthur Samuel is that there are two essentially different computational strategies for assessing the values of actions. One (called model-based control) involves building a search tree of possible options and outcomes. This imposes heavy demands on computation but turns out to be easy to learn. The other (model-free) involves learning how to make predictions about long-run consequences by enforcing consistency between successive estimates. This eliminates errors such as switching from thinking that things are going well to going badly without any apparent reason. Since the predictions are immediately available, without any need to build or evaluate a tree, model-free control is computationally straightforward; but learning is harder.

That these systems operate at different sweet spots of learning and computation implies that having both can be useful. Indeed, there is by now ample psycholog-



ical evidence that animals and humans do so, and quite some work on the underlying neural systems. By understanding how factors such as differential uncertainties (another notable jewel in our gilded age) influence how their outputs are integrated, we can see how to achieve intelligently adaptive decision-making.

A second primary insight is the importance of nature as well as nurture, i.e., the extensive (evolutionary or Pavlovian) pre-programming about decision-making that we and other animals enjoy. Although these can lead to a panoply of choice 'illusions' when current circumstances vary from ingrained prior assumptions, it is increasingly obvious that there is a wonderfully sophisticated scheme for assessing and reacting to opportunities and threats, which dramatically reduces the load for individual learning. We are just starting to have the tools to understand the nature and scope of this preprogramming, to fathom its mix of fixedness and adaptability, and to realize it in artificial systems.

The blistering pace of development in the area as a whole means that in the next decade or two we can look forward to systems that learn from their experiences to make smart informed decisions about the future. These kinds of systems, inspired by the workings of the brain, will be embedded in every facet of our lives from healthcare to finance to entertainment. These advances will likely render the errant babble of the Loebner prize a noisy sideshow on the path towards true human-level artificial intelligence. **d**

Peter Dayan & Demis Hassabi

Peter Dayan is the director of the Gatsby Computational Neuroscience Unit at the University College London. He is the co-author of 'Theoretical Neuroscience', a leading textbook of computational and mathematical modelling of neural systems. He began his career studying Mathematics at Cambridge followed by a PhD in Artificial Intelligence at Edinburgh. He did his post doctorate at the Massachusetts Institute of Technology in Boston. Demis Hassabi is one of his post-doctoral students.